Flutter Margin Computation

AEROELASTICITY

TEAM 05

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1 Introduction

The objective of this study is to estimate the critical flutter velocity of an aircraft by extrapolating data gathered during subcritical flight regimes.

Flight Vibration Testing (FVT) is a critical phase in the certification of any aircraft, requiring a balance between expanding the flight envelope and ensuring safety against aeroelastic instability.

Flutter is a dynamic aeroelastic instability that involves the coupling of two or more structural modes. It arises from the interaction of aerodynamic, elastic, and inertial forces, resulting in self-excited, divergent oscillations that, if not reverted, can lead to catastrophic structural failure.

Flutter numerical analysis of the first prototype of the aircraft, has revealed that the critical flutter mechanism is a classical binary flutter driven by the coupling of wing bending and wing torsion modes. The average frequencies of the first flight flutter test runs are shown in Figure 1.

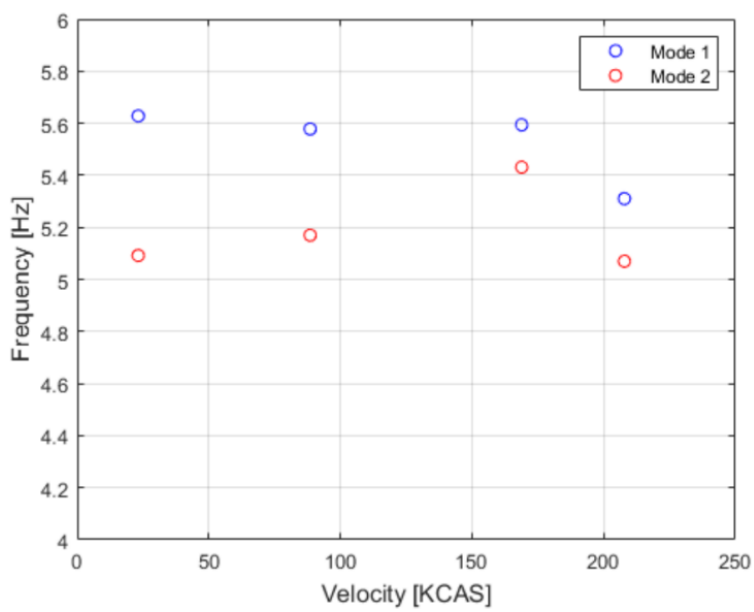

Figure 1: Average frequencies of the first flight flutter test runs

The oscillations generated are usually damped when at low speeds. Structural failure occurs at high speeds, above the flutter speed. For the reference damping, the group number should be used as following: this is Group 5, so $G_1 = 5 \cdot 10^{-3} = 0.005$. Note that the provided damping figures are positive, so $g > 0$ will be considered positive, unlike what the class slides state [1].

In order to solve this problem, a damping evaluation analysis will be carried out using both a linear and a 4th order extrapolation and then implementing the flutter margin equations that will be explained below.

2 Paper review

The paper *Zimmerman, N.H. And Weissenburger, J.T., "Prediction of Flutter Onset Speed Based on Flight Testing at Subcritical Speeds", Journal of Aircraft, Vol. 1 No. 4, pages 190-202. July-August 1964* [3] has been used for the analysis of the flutter margin. In this paper formulation, the rate of decay is given by:

$$\beta = \omega \cdot \zeta \tag{1}$$

The expression for the Evaluation of Flutter Margin from Flight Data is the following:

$$\begin{aligned} F = \left[\left(\frac{\omega_2^2 - \omega_1^2}{2}\right) + \left(\frac{\beta_2^2 - \beta_1^2}{2}\right)\right]^2 \\ + 4\beta_1\beta_2 \left[\left[\left(\frac{\omega_2^2 + \omega_1^2}{2}\right) + 2\left(\frac{\beta_2 + \beta_1}{2}\right)\right]^2 \right. \\ \left. - \left[\left(\frac{\beta_2 - \beta_1}{\beta_2 + \beta_1}\right)\left(\frac{\omega_2^2 - \omega_1^2}{2}\right) + 2\left(\frac{\beta_2 + \beta_1}{2}\right)\right]^2\right] \end{aligned} \tag{2}$$

The Flutter Margin Prediction Equation is expressed as:

$$F = B_2(C_{L_\alpha} q)^2 + B_1(C_{L_\alpha} q) + B_0 \tag{3}$$

In equation 3, B_i are coefficients where configuration constants from previous steps are embodied (not shown in this report).

A particularly convenient expression of the flutter prediction equation can be developed by considering C_{L_α} as a constant, so it will be introduced inside the "constant" term (expressed in equation 4 in brackets). In this way, equation 3 becomes the following:

$$F = (B_2 C_{L_\alpha}^2) q^2 + (B_1 C_{L_\alpha}) q + B_0 \tag{4}$$

And this yields:

$$\boxed{F = B_2' q^2 + B_1' q + B_0'} \tag{5}$$

where $B_i' = B_i (C_{L_\alpha})^i =$ constant. This means that the flutter margin varies parabolically with q. Equation 5 is nearly correct for predicting Flutter Margin, except for large variations of C_{L_α}.

However, considering that the expression for q is given by:

$$q = \frac{1}{2}\rho V^2 \tag{6}$$

Substituting expression 6 in equation 5, and including the constant value of $\frac{1}{2}\rho V^2$ inside the

constant in the same way as the C_{L_α} before, it finally yields:

$$\boxed{F = B_2'' V^4 + B_1'' V^2 + B_0''} \tag{7}$$

where, recalling all the simplifications, $B_i'' = B_i'(\frac{1}{2}\rho)^i (C_{L_\alpha})^i = \text{const.}$

The flutter margin, F, is a critical parameter in the study of the flutter phenomena, so a safety margin between the flutter speed and the maximum design speed needs to be established. These values must not be negative, since this will mean the aircraft is prone to flutter.

3 Direct damping evolution with flight speed

In the study of flutter margin, data shown in figure 1 is to be analyzed with two extrapolation techniques based on polynomial fitting: a linear fitting (1st order polynomial) and a 4th order fitting curve.

From figure 1, the following data is extracted:

Velocity [KCAS]	**Mode 1 frequency [Hz]**	**Mode 2 frequency [Hz]**
∼23	5.63	5.10
∼88	5.58	5.17
∼170	5.60	5.43
∼208	5.31	5.07

Table 1: Data extracted from figure 1

From [1], it is known that when the damping curve of a mode crosses the value of $g = 0.00$ line, the flutter speed, V_f, is found at that specific point.

The speed separation between $g = 0.00$ and $g = 0.03$, indicates how severe the flutter is:

$$\Delta V = V_{g=-0.03} - V_{g=0.00} \tag{8}$$

- $\Delta V <\sim 10$ KEAS $\implies$ ABRUPT flutter
- ~ 10 KEAS $< \Delta V <\sim 50$KEAS $\implies$ SEVERE flutter
- ~ 50KEAS $< \Delta V <\sim 100$KEAS $\implies$ MODERATE flutter
- ~ 100 KEAS $< \Delta V <\sim 200$KEAS $\implies$ LIGHT flutter
- $\Delta V >\sim 200$ KEAS $\implies$ MILD flutter.

If the curve never crosses $g = 0.03$, the flutter is of concern, but it can be considered as BENIGN flutter.

If the fitted curve never goes below $g = 0.01$, the resulting flutter is allowed by regulations and can be considered as another mode, called "hump mode".

4 Results

4.1 Flutter speed prediction through Direct Damping Evolution

Figure 2, presents a scatter plot with the provided measurements for the evolution of the Structural Damping (g) vs Flight speed [$KCAS$]. Velocity and frequency data were extracted from Figure 1, and along with the values on the table on the following slides, the figure is constructed.

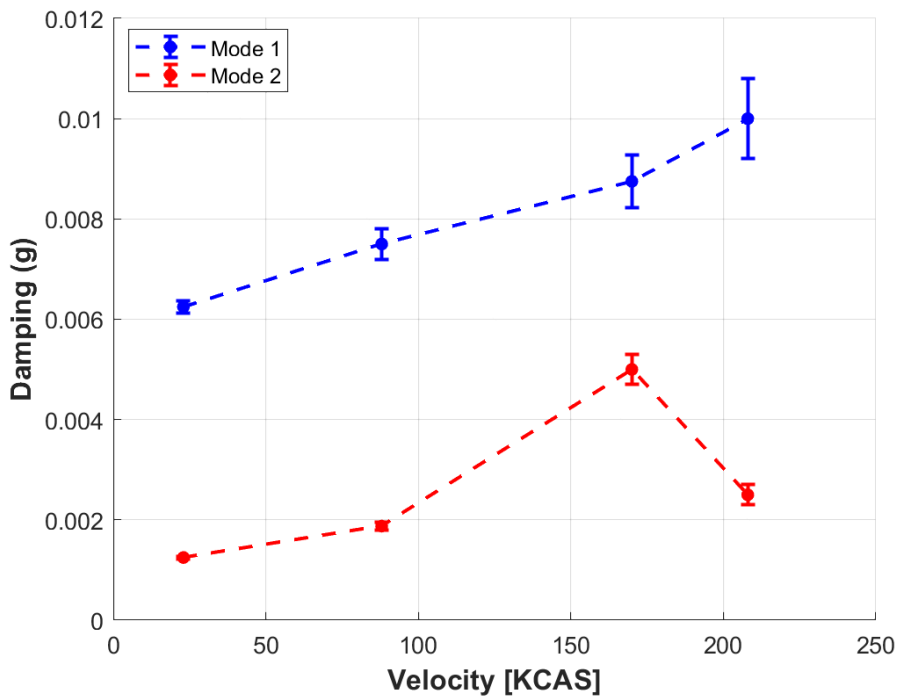

Figure 2: Structural Damping (g) vs Flight Speed [KCAS]

Mode 1, represented in blue, is the least critical mode, indeed it is not relevant at all for the current analysis since its damping values increase monotonically. Therefore its velocity will never reach the zero g line, not having any flutter speed, V_f. The analysis will be only based on Mode 2, which is the most critical mode because it presents a decreasing trend, meaning that, in this case, the structural damping value for does approach zero, indicating the existence of a flutter speed V_f.

As mentioned before, both a linear and a 4th degree extrapolation fit have been used to do the study. An important aspect to point out is that 3 curves have been obtained for each fit, corresponding to the optimistic, mean and pessimistic cases. The optimistic and pessimistic curves have been obtained, respectively, by the upper and lower bounds of the error bars belonging to the mean value curves.

Another aspect to explain is the equivalence between KCAS and KEAS values. At sea level conditions, these velocities are identical by definition, while at other altitudes are theoretically

different. However, for subsonic velocities, the numerical differences between KCAS and KEAS remain minimal too. So when performing the data analysis for the severity of the flutter, KCAS and KEAS are equivalent in magnitude for all the cases.

After having considered all the aspects mentioned until this point, the analysis can be done.

4.1.1 Linear fit

Figure 3 presents the linear fitting of Mode 2's damping in which only the last two data points have been used to provide the estimation. This approach has been considered because only with this two points, the extrapolation will cross the $g = 0.00$ line, giving the flutter speed V_f.

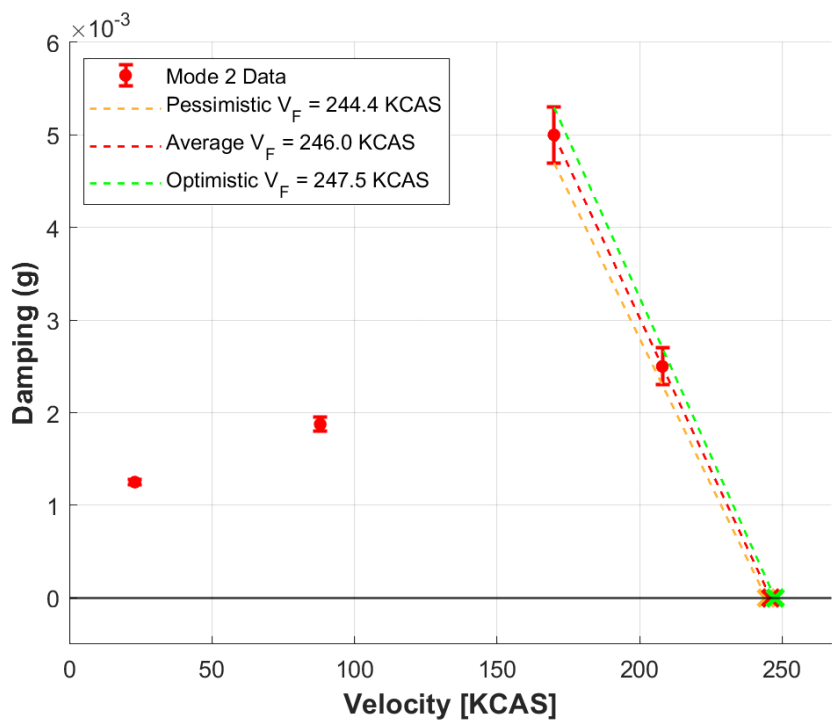

Figure 3: Linear fit to predict flutter speed

It is known from the theory that the flutter speed is given when $g = 0$. If figure 3 is zoomed in that value:

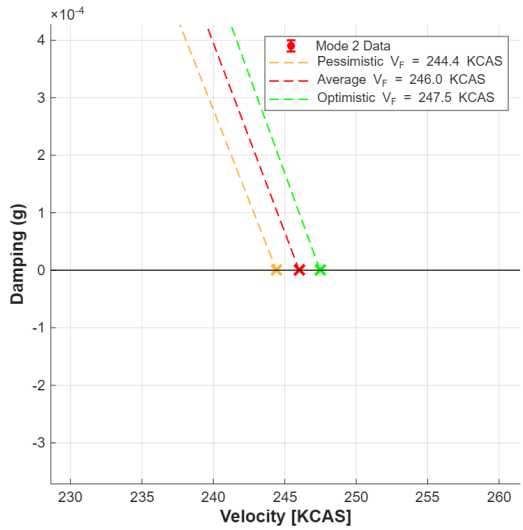

Figure 4: Linear fit zoomed in $g = 0$

By analyzing figure 4, it can be seen that the flutter speed interval of the linear fit is:

$$V_f = [244.4,\ 247.5] \text{ KCAS}$$

With a mean value of:

$$\boxed{V_f = 245.95 \text{ KCAS}}$$

The results for ΔV can be graphically shown in figure 5:

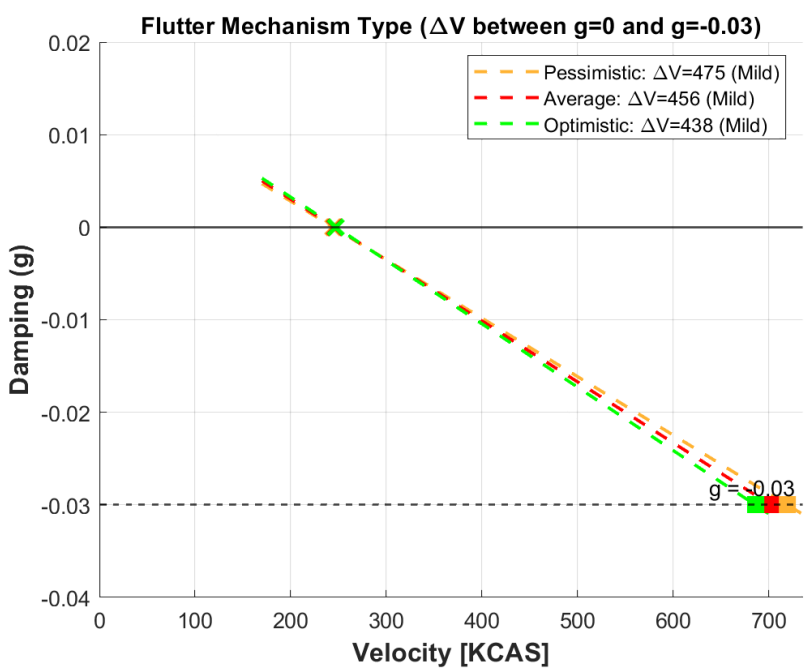

Figure 5: ΔV_{max} for the 3 cases studied (linear fit)

Moreover, in section 3, a scale of the severity of the flutter mechanism in relation with the ΔV has been established. The data from figure 5 can be analyzed as follows:

- Case 1: Pessimistic V_f:

$$\Delta V_{max} = 475 \text{ KCAS} \sim 475 \text{ KEAS} > 10 \text{ KEAS} \implies \text{MILD flutter}$$

- Case 2: Average V_f:

$$\Delta V_{max} = 456 \text{ KCAS} \sim 456 \text{ KEAS} > 10 \text{ KEAS} \implies \text{MILD flutter}$$

- Case 3: Optimistic V_f:

$$\Delta V_{max} = 438 \text{ KCAS} \sim 438 \text{ KEAS} > 10 \text{ KEAS} \implies \text{MILD flutter}$$

4.1.2 4th order polynomial fit

From the theory slides, it is known that the damping value must start at 0, meaning that it must have a 0 value at the origin as well. Considering all the data points available, the 4th degree polynomial interpolation has been found to be the most suitable extrapolation method for performing this study. This approach is depicted in Figure 6, which yields a less optimistic, but more realistic prediction for the flutter speed V_f, with both a reduced value and lower data dispersion compared to the linear approach.

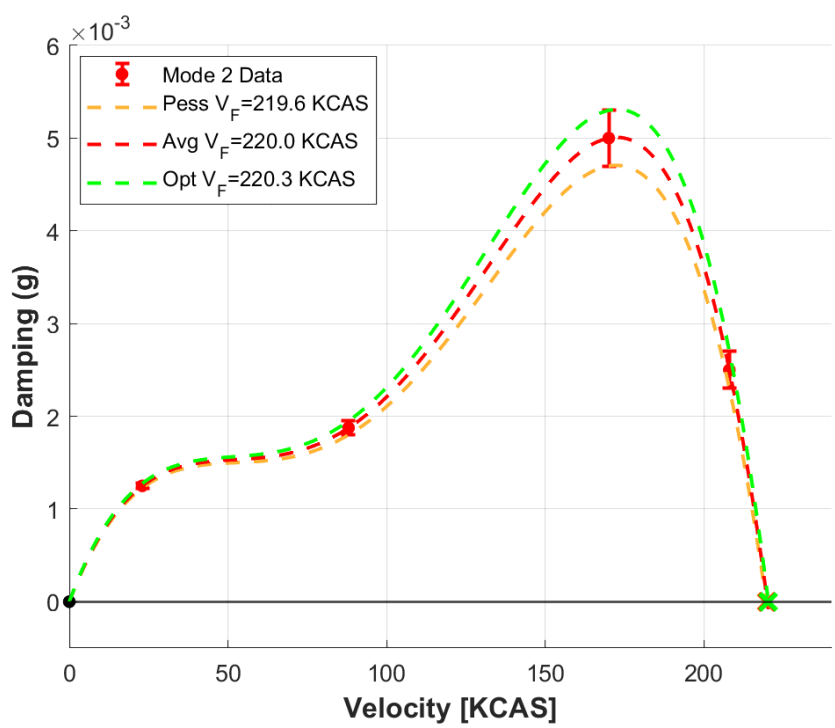

Figure 6: 4th order fit curve to predict flutter speed

Zooming figure 6 at $g = 0$:

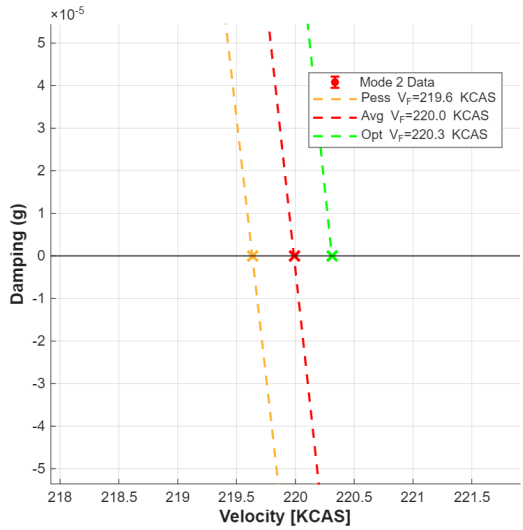

Figure 7: 4th order fit zoomed in $g = 0$

By analyzing figure 7, it can be seen that the flutter speed interval of the 4th order fit is:

$$V_f = [219.6,\ 220.3] \text{ KCAS}$$

With a mean value of:

$$\boxed{V_f = 219.95 \text{ KCAS}}$$

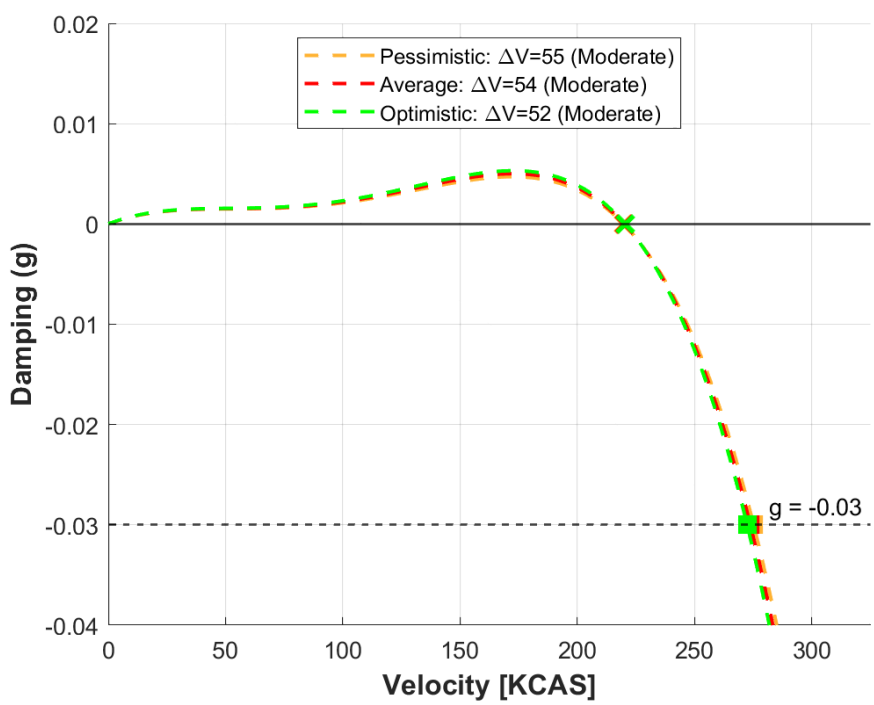

Figure 8: ΔV_{max} for the 3 cases studied (4th order polynomial fit)

By looking at the values shown in figure 8 and following the severity scale of the flutter with respect to the ΔV, already mentioned in section 3, the following analysis has been carried out:

- Case 1: Pessimistic V_f:

$$\Delta V_{max} = 55 \text{ KCAS} \sim 55 \text{ KEAS} \implies \sim 50 \text{ KEAS} < \Delta V_{max} <\sim 100 \text{ KEAS} \implies \text{MODERATE flutter}$$

- Case 2: Average V_f:

$$\Delta V_{max} = 54 \text{ KCAS} \sim 54 \text{ KEAS} \implies \sim 50 \text{ KEAS} < \Delta V_{max} <\sim 100 \text{ KEAS} \implies \text{MODERATE flutter}$$

- Case 3: Optimistic V_f:

$$\Delta V_{max} = 52 \text{ KCAS} \sim 52 \text{ KEAS} \implies \sim 50 \text{ KEAS} < \Delta V_{max} <\sim 100 \text{ KEAS} \implies \text{MODERATE flutter}$$

4.2 Flutter speed prediction through Flutter Margin equation

In this section, the flutter speed is to be predicted with a different approach, using equation 7. From the 4th order fitting, the following figure is obtained, where now the flutter margin is plotted against the velocity squared:

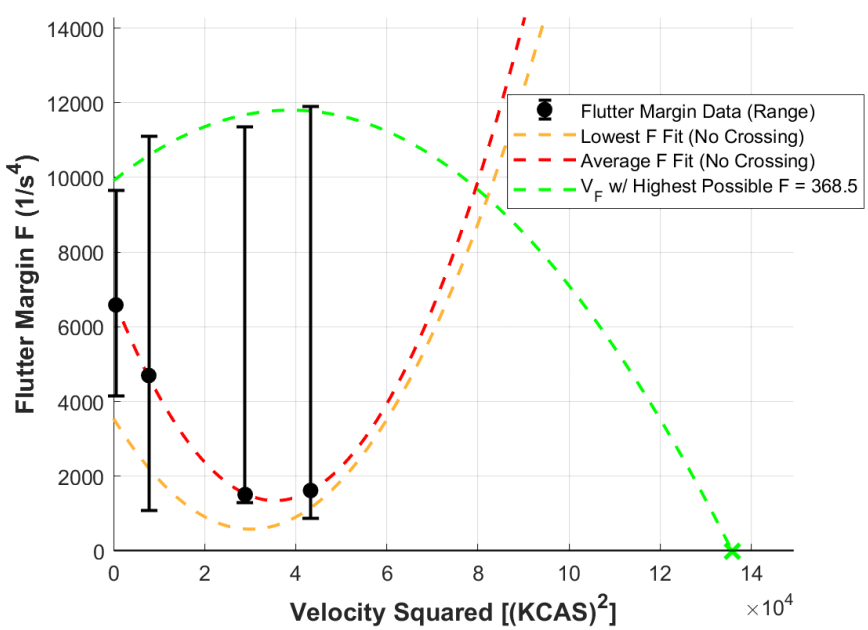

Figure 9: Flutter Margin Maximum & Minimum

The figure above presents a rare behavior in which the most pessimistic scenario for flutter margin, nor the average one gives flutter. Using the lowest values, the curve does not intersect the zero flutter margin axis, indeed it presents an increasing behavior for increasing speeds.

The exact same phenomena happens when considering the average flutter margin fitting, it behaves as a convex parabola, never crossing zero and increasing for large velocities. Finally, when looking at the most optimistic scenario, the curve intersects zero in a rarely high velocity, and as this curves totally differs from the two other ones, it might not be accurate enough to take it as definitive.

In summary, each combination produces a totally different curve, leading to an infinite range of potential results. This high discrepancies indicate that the use of the simple *optimistic*0 or *pessimistic* scenario based on maximum or minimum values is not enough to obtain the flutter speed, as it can produce radically different outcomes.

A different approach is proposed to deal with this high discrepancies in the results: carrying out a Monte Carlo simulation to explore all possible combinations of damping and frequency.

In this numerical procedure, the experiment is ran several times (500,000 in this specific case), and the predicted flutter speed frequency is annotated. Then it is plotted in a Probability Density plot (an histogram). The scatter of results is assumed to follow a normal distribution, which is later consistent wiht the results obtained. It generates a mean curve and a confidence interval (90% interval), thereby providing a more robust estimation of flutter speed than simply considering the extremes, since very low frequencies are obtained for values on the extremes of the flutter speed range (0.1% of simulations predicted $V_f > 500$ KCAS, similarly happens with $V_f < 100$ KCAS).

Moreover, the scatter is quantified as twice the standard deviation (2σ), corresponding to a 95% confidence interval under normal distribution assumptions, following standard uncertainty analysis protocols in engineering simulations dictated by ISO [2]: for the estimation of uncertainty in engineering, a probabilistic extrapolation of the results can be done to obtain the flutter speed, but due to the criticality of the flutter phenomena it must have a reasonably high confidence interval, for this case a 95% confidence interval is coherent.

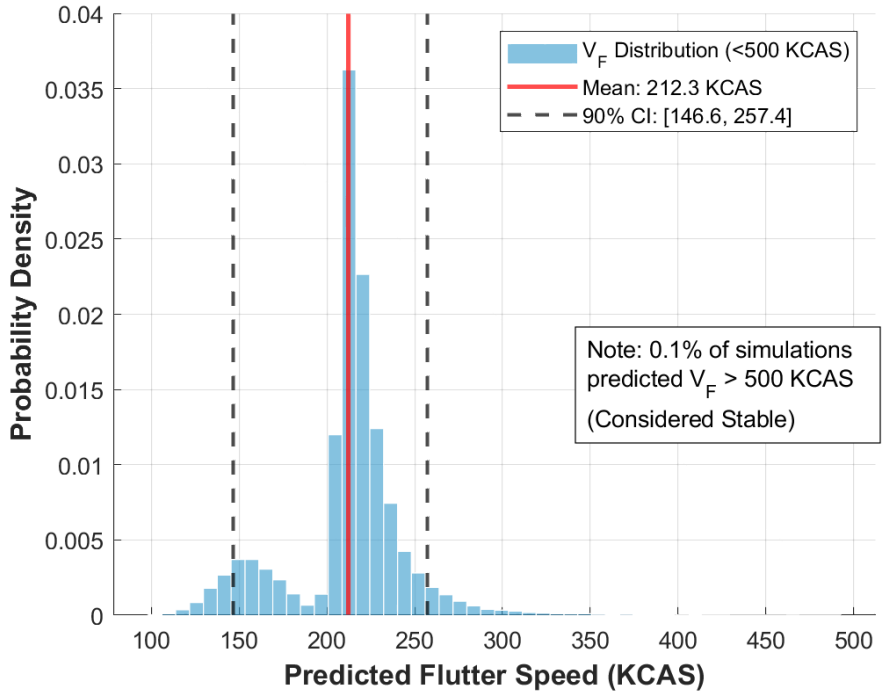

Figure 10: Monte Carlo Simulation for predicting Flutter Speed

From the results of the simulation shown above, the flutter speed is selected to be the mean value:

$$\boxed{V_f = 212.3 \text{ KCAS}}$$

4.3 Strategy decision

After having carried out both the direct damping evolution analysis and the Monte Carlo simulation for the flutter margin method, the next step is to decide the strategy for the next phase of the Flight Vibration Test.

For the first method for computing the flutter speed, the linear fit was proven no to be a proper approximation. Whereas the 4th order polynomial fitting is coherent with the four data points. A recommended approach would be to continue with performing more flight tests to cover more data points and obtain a much more physically accurate result for the fitting of this data and the consequent accuracy of the flutter speed.

The Monte Carlo analysis returns a statistically robust estimation of the flutter speed, with most of the values well within the 95% confidence interval, and with very few extreme values. The scatter of the expected flutter speeds is properly modeled.

For the next phase, it is advisable to plan at least five to ten more additional flight points, with small speed increments of around 10–20 KCAS for speeds between 150-208 KCAS, since 150 is approximately the lower bound of the 95% confidence interval, and 208KCAS is the highest velocity obtained from 2. It would be ideal to have more data points for velocities up

to this one in order to reduce uncertainty and scatter of results, so that finally the predicted flutter speed is more accurate when using flutter margin approach.

Reaching speeds larger than 208 KCAS is too risky since it could easily reach the estimated flutter speed, 212.3 KCAS. Obtaining more results below the mean value, also yields into the predicted flutter speed. More data points entail lower scatter and uncertainty, and thus the flutter speed estimation through the flutter margin prediction equation will be more accurate. If this better estimate turns out to be significantly higher than the 208 KCAS (218 KCAS or more), a flight test around 210 KCAS could be performed to improve the accuracy of the predicted V_F

Finally, the numerical simulation gives no reason to abort the current test campaign, since almost all the results for the predicted flutter speed are within reasonable flight test speed, as seen in 4.2.

References

[1] Tamara Casillas, Héctor Climent, and Pablo Fajardo. *Aeroelasticity. Lecture 07. The Flutter Equation and Solution*. Lecture Slides. From the course "Aeroelasticity" in the Degree in Aerospace Engineering. 2025.

[2] International Organization for Standardization. *ISO Guide 98-3: Uncertainty of Measurement – Part 3: Guide to the Expression of Uncertainty in Measurement (GUM:1995)*. Available from ISO. 2008.

[3] Norman H. Zimmerman and Jason T. Weissenburger. “Prediction of Flutter Onset Speed Based on Flight Testing at Subcritical Speeds”. In: *Journal of Aircraft* 1.4 (1964), pp. 190–202.